Agricultural production

Traditional crops offer a taste of tomorrow

The human diet of the future can draw on traditions from regions that share similar climate conditions — but farmers need more security to experiment

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Mixed cultivation: longans, dragon fruits, durians and jackfruit trees growing in Vietnam © Magdalena Chodownik/Anadolu via Getty Images

Innovations in storage and the rise of global supply chains have made a wide variety of fruits, grains and vegetables available or even commonplace, often year-round, for millions of people across the globe. But that is set to change as rising temperatures and erratic weather patterns put pressure on crop yields.

Annual agricultural losses related to natural disasters have more than doubled in three decades, according to the UN's Food and Agriculture Organization, rising to an average annual cost of $144bn in recent years. Disruptions to the agricultural, food and beverage industries from drought-related extreme weather events have risen by 250 per cent since 2021, according to supply chain tracker Resilinc.

“The food system we have is not going to be fit for purpose in 30 years,” says Jessica Fanzo, a professor of food and climate at Johns Hopkins University’s School of Advanced International Studies in Italy. “It’s going to be really hard to raise livestock [and] it’s going to be very really hard to grow maize and soyabeans in Brazil,” one of the world’s top exporters of the two staple crops, she adds.

According to the FAO, just nine main species accounted for nearly 70 per cent of global crop production for human consumption by 2014, the agency said in 2019. Reversing that trend would help slow the spread of pests and crop diseases and build a more sustainable agricultural system.

That would mean revitalising some of the several thousands of traditional crops that have been neglected in favour of monocrops — where farmers grow only one type over a large area — such as maize or rice, as well as experimenting with those that are more climate-resilient. Experts in climate and nutrition say such strategies could lead to consumers having access to a wider variety of foods, and being able to hold on to more foods that hold particular cultural significance for them.

But that would require making it more financially manageable for the farmers growing these foods to experiment with crops, and for them to access the necessary knowledge, traditional or modern, says Fanzo.

Tomorrow’s crops: balancing old and new



Maturing sorghum in Jiangsu province, China © Cfoto/Future Publishing via Getty Images

International organisations, from the European Commission, executive arm of the EU, to the FAO, recognise the role that traditional crops could play in helping to feed growing populations as the climate changes. This category includes thousands of plants such as: manoomin, a wild rice native to North America; sorghum, the drought-hardy grain that is a dietary staple in north and sub-Saharan Africa and south Asia; or jackfruit, a tropical fruit that is increasingly used as a plant-based alternative to meat.

Interest in revitalising traditional crops has grown over several decades. In India, a flagship programme that aims to promote traditional varieties of millet has helped increase production of the grain to 18mn tonnes in 2025.

Beyond their benefits in biodiversity and climate resilience, these crops often provide significant nutritional benefits and carry cultural significance, says Parvinder Kumar, founder of Wayfield Agricultural Technology, an agricultural technology company in northern India.

"[Traditional crops] are connected to farmers' family histories, local food cultures, community identities, and knowledge passed down through generations," he says. "Farmers often understand these crops very well, including how to grow, harvest, store and prepare them."

But the farmers who want to grow these grains face many hurdles, he says. The financial return is often too low, while shifting climate conditions are making it harder to predict the yields.

One answer to such challenges is to connect farmers to speciality food and ingredient markets that will pay higher prices. In the region Kumar works in — the Indian side of the Himalayan mountains — black soyabean grows well, but it is less popular as an ingredient in India than in Japan or South Korea. Connecting the smallholder farmers to Japanese and Korean restaurants in the region helped increase growers' income from the crop.

Finding ways to increase crop yield is also crucial, says Kumar. Sometimes that involves integrating regenerative practices, such as intercropping where crops grow alongside each other. For instance, yield from the black soyabean can go up by around 5 per cent when planted with finger millet, says Kumar.

Diversification of crops benefits these farmers' businesses, he adds. Introducing new crops such as apples, he says, plays a dual role. Not only do they sell for more on the market than traditional staples, but fruit trees can help mitigate soil erosion.

"However, introducing new crops requires careful assessment," he adds.

The need for knowledge transfer

“Agricultural knowledge is often fragmented,” Kumar says. Many young people are leaving farming in search of a more stable and profitable profession — as Kumar himself did when he left agriculture to study technology before returning to his family trade. In addition, it is often difficult to access technical or scientific information if it is unavailable in local languages.

The prickly pear is one example of a successful knowledge transfer. The cactus plant is native to Mexico, where it is used in tacos, candy or sweets, or as a juice. It grows in many arid regions, including Tunisia and Madagascar — and for a long time, it was considered invasive.



Prickly pear and onion taco, served in Mexico, where the cactus plant is a popular ingredient © Omar Torres/AFP via Getty Images

“If you don’t have that cultural knowledge of how to prepare it, it’s the most hostile food to prepare,” says Quinault Childs, a former research director of the Food Futures Lab at the Institute for the Future, a California-based research organisation. “You’ve got spines in your hands, and you’re, like, ‘what do you do with all this?’”

In 2021, when drought threatened several areas where the prickly pear grows well, the FAO published a report detailing how the crop can be used to help restore soil health and to help prevent famine when all other crops fail. That effort spread knowledge about the use of prickly pear, says Childs.

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Quinault Childs

"There's a huge, huge, huge, need for knowledge transfer," he says. "There's so much latent knowledge in food production systems that is outside of western conventional agriculture . . . but that doesn't necessarily scale up to a farmer in Iowa. It doesn't get codified and it doesn't get transferred."

Technological change in crop production "is going to happen [anyway] and the world is really set up to do that well", he says.

"What the world is a little bit less well-suited to do right now is to . . . respect and spread all of those different little niche pockets of knowledge that together could make a huge difference in how we produce and consume and interact with food."

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